Vmem-φ: Zero-Cost Out-of-Distribution Detection in Spiking Neural Networks from Membrane-Potential Statistics

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Abstract

Spiking neural networks (SNNs) on event cameras are deployed precisely where the standard out-of-distribution (OOD) toolkit cannot run: on neuromorphic silicon with no floating-point logits, no gradients, and no budget for a second forward pass. We observe that the sub-threshold membrane potential $V_{\mathrm{mem}}(t)$ of the Parametric Leaky-Integrate-and-Fire (PLIF) neurons is a free byproduct of the forward pass – a register the substrate already maintains – and that its per-channel statistics $\varphi = [\mu, \sigma^2, \kappa]$ (a 2112-D descriptor over 704 channels $\times$ 4 layers) form an OOD signal at zero added inference compute – a register read plus three moments, with no second forward pass – on which a lightweight host-side detector then operates. We contribute Gen1-C, a reproducible, seeded event-camera corruption benchmark of six sensor-grounded corruptions $\times$ five severities – each traced to a concrete hardware failure mechanism – plus a released membrane-statistics feature set so OOD detectors can be compared without re-running SNN inference. We give a closed-form leaky-integrator theory that predicts, per corruption, which membrane moment moves and therefore which corruptions are detectable and which are provably not, and we confirm with a motivation experiment that feeding these corruptions to a pretrained state-of-the-art hybrid SNN–ANN detector (no retraining) collapses its mAP. Finally we introduce the Manifold-Decomposition Detector (MDD), which decomposes each frame's deviation into orthogonal manifold axes scored two-sided, taking the contraction corruptions above chance for the first time; one unsupervised, corruption-blind score reaches ≥ 0.95 AUROC on four of six corruptions with sequence aggregation (two of them 1.00). We report honest negatives – polarity_flip (a network invariance) and spatial_dropout (a signature global pooling discards) – and frame them as an open challenge: the Achilles' heel of event-based perception.

1 Introduction

Event cameras report per-pixel brightness changes asynchronously at microsecond resolution, high dynamic range, and low power; paired with spiking neural networks (SNNs) they form the canonical low-power perception stack for neuromorphic hardware. As these systems move toward safety-critical deployment (automotive, robotics), a basic question becomes unavoidable: when the sensor input drifts away from the training distribution, can the system notice? Out-of-distribution (OOD) detection is the standard answer for artificial neural networks (ANNs), but essentially every method in that literature assumes a capability the neuromorphic execution model does not provide – floating-point logits (MSP, Energy), input gradients (ODIN, GradNorm), or a second forward pass over the features (ReAct, ANN-feature Mahalanobis/kNN). On a chip that emits spikes and integer events under a strict energy budget, these are not merely expensive but architecturally infeasible (Section 3.5).

We start from a signal that is free exactly where the alternatives are impossible. The Parametric Leaky-Integrate-and-Fire (PLIF) neuron maintains a sub-threshold membrane potential $V_{\text{mem}}(t)$ – physical state the substrate already updates on every time step to decide whether to spike. Reading it and reducing it to a few per-channel moments adds zero multiply–accumulates and zero weight fetches. We call the resulting 2112-dimensional descriptor φ and ask how far this free signal goes as an OOD detector against realistic event-camera corruptions, each grounded in a concrete sensor failure mechanism (Figure 1).

Contributions. We make three key contributions.

1. A zero-cost membrane OOD signal and the Gen1-C benchmark. We introduce φ, an OOD signal harvested from the PLIF membrane at zero additional compute, and argue it is the richest OOD signal available at zero added inference cost on the target neuromorphic hardware – needing no logits, gradients, or second forward pass, and provably subsuming spike-rate signals (Section 3.5). To evaluate it we release Gen1-C: a reproducible (seeded, deterministic, model-agnostic) event-camera corruption benchmark of six corruptions × five monotonic severities, each mapped to a physical hardware failure mode (Section 3.2, Figure 1), in the ImageNet-C (Hendrycks & Dietterich, 2019) tradition, plus a released membrane-statistics feature set.
2. A closed-form theory of detectability, empirically motivated. We derive, from the PLIF state equation and with no parameters fit to corruption labels, a leaky-integrator theory that predicts which membrane moment each corruption moves and therefore which corruptions are detectable – a dilation/contraction/invariant taxonomy (Section 3.3). We further prove the membrane mean, variance, and the Mahalanobis OOD score all increase monotonically with corruption severity (Theorems 1–3), and that the sub-threshold potential strictly subsumes spike counts for OOD (Proposition 1). We confirm the stakes with a motivation experiment: a pretrained SOTA hybrid SNN–ANN detector (Neftci et al., 2025) loses substantial mAP under these corruptions with no retraining (Section 4.2).
3. The Manifold-Decomposition Detector (MDD), and an open challenge. Following the geometric diagnosis, MDD stops collapsing φ to one scalar and instead scores orthogonal, two-sided manifold axes fused by a calibrated OR (Section 3.6); it takes the contraction corruptions above chance for the first time and reaches $\geq$ 0.95 AUROC on four of six corruptions with sequence aggregation. We report polarity_flip and spatial_dropout as honest, theory-predicted residuals and release them as a community challenge (Section 5).

2 Background and Related Work

OOD detection for ANNs. The dominant families score a post-hoc statistic of a trained network: maximum softmax probability (Hendrycks & Gimpel, 2017), temperature-scaled and input-perturbed logits (Liang et al., 2018), free energy (Liu et al., 2020), rectified activations (Sun et al., 2021), virtual-logit matching (Wang et al., 2022), gradient norms (Huang et al., 2021), and distance/density on penultimate features (Mahalanobis (Lee et al., 2018), deep k-NN (Sun et al., 2022)). All presuppose floating-point readouts, gradients, or repeated forward passes – capabilities absent from the spiking, feed-forward, integer-event execution model of neuromorphic hardware (Section 3.5 makes this precise).

OOD detection for SNNs (and how we differ). OOD detection inside an SNN is not unprecedented. Most directly, Martínez et al. (Martínez et al., 2023) characterise the internal activations of hidden layers via spike-count patterns and add an explainability map, evaluated on image (including event-based) classification benchmarks. Our work is distinct on four axes that, in combination, define a new task: (i) we read the sub-threshold membrane potential, not spike counts – a continuous, pre-threshold signal we prove strictly subsumes the binarised spikes for OOD (Proposition 1: spike counts are blind to sub-threshold shift); (ii) the host model is an event-camera object detector (Prophesee Gen1), not a classifier, so there is no class posterior to lean on; (iii) the distribution shifts are event-stream corruptions we define and release as a benchmark, not held-out classes; and (iv) we make the zero-compute / neuromorphic-register feasibility argument central. We therefore position the paper as introducing the event-camera membrane-statistics OOD task and the Gen1-C benchmark, while explicitly crediting prior SNN-OOD work.

Event-camera robustness and corruption benchmarks. Robustness benchmarks built from synthetic corruptions are standard for frame cameras (ImageNet-C (Hendrycks & Dietterich, 2019)); the event-camera analogue is comparatively underdeveloped. Gen1-C (Section 3.2) fills this gap with corruptions grounded in concrete sensor failure modes (Figure 1, Table 1), seeded for exact reproducibility, and operating on the standard histogram tensor so they compose with any event model.

The host detector. Our membrane signal is read from the pretrained hybrid SNN–ANN detector of Neftci et al. (Neftci et al., 2025), a strong Gen1 baseline that fuses a spiking front-end (SpikingJelly PLIF (Fang et al., 2021)) with an ANN detection head. We treat it as read-only and never retrain it – both for the OOD signal and for the motivation experiment of Section 4.2.

Membrane potential as a hardware register. On Loihi (Davies et al., 2018), BrainScaleS (Schemmel et al., 2010), and SpiNNaker (Furber et al., 2014), V_{mem} is state the neuron already maintains; harvesting φ from it is an $O(\text{channels})$ read. This is the basis of the feasibility argument in Section 3.5.

3 Method

We present (i) the zero-cost membrane descriptor φ, (ii) the Gen1-C corruption suite with its physical-mechanism grounding and algorithms, (iii) a leaky-integrator theory that predicts per-corruption detectability, (iv) the neuromorphic-hardware feasibility argument, and (v) the Manifold-Decomposition Detector that follows from the theory.

3.1 The zero-cost membrane descriptor φ

The membrane signal is read from the pretrained hybrid SNN–ANN detector with a forward hook on all four PLIF layers (704 channels total). Concretely, V is the neuron's membrane state module.v at each of the T=10 unrolled steps (reset-inclusive: we read the raw maintained register, we do not excise the steps around a spike); the linear sub-threshold analysis of Section 3.3 is therefore an approximation whose reset effects enter as a bounded error term (Theorem 1), exact in the low-firing regime. For each channel c we apply Global Average Pooling (GAP) over the spatial dimensions of V and compute its first three central moments across the T steps,

$$\varphi_c = \big[\mu_c, \sigma_c^2, \kappa_c\big], \qquad \mu_c = \mathbb{E}[V_c], \quad \sigma_c^2 = \mathrm{Var}[V_c], \quad \kappa_c = \frac{\mathbb{E}[(V_c-\mu_c)^4]}{\sigma_c^4}, \tag{1}$$

stacking to a 2112-D descriptor φ per frame. The mean captures sustained drive, the variance fluctuation energy, the kurtosis burstiness. Two properties drive every later result: (i) GAP discards spatial layout – φ knows how much each channel fired, not where; (ii) φ inherits the network's invariances. A spatial-dispersion extension phi_spatial (1408-D: per-channel spatial variance and participation ratio) recovers part of the structure GAP discards. Clean φ lies on a thin, curved, scene-dependent manifold; detection is deciding whether a new φ lies on it.

3.2 The Gen1-C event-camera corruption suite

A benchmark for OOD detection is only as meaningful as the distribution shifts it tests against. We define a controlled, reproducible battery of six corruptions that model the dominant failure modes of event-camera front-ends, each at five monotonically increasing severities; with the clean baseline this yields $6 \times 5 + 1 = 31$ runs. Each corruption perturbs a structurally distinct axis of the input – DC offset, localized burst, temporal order, global rate, polarity, or spatial support – which is what makes the closed-form analysis of Section 3.3 possible.

Input representation. Every corruption operates on the model's input tensor, so it composes with any downstream architecture and is label-free. A sequence is a stack of N event-histogram frames of shape $(20, H, W)$, $H \times W = 240 \times 304$ for Gen1. The 20 channels are two polarities $\times$ ten $50\,\mu$s time bins: channels 0–9 ON (oldest to newest), 10–19 OFF. Counts are uint8; corruptions promote to a wider type internally and clip to $[0, 255]$.

Design principles. (i) Reproducibility: every random decision is drawn from a single seeded NumPy generator, so a (corruption, severity, seed) triple is deterministic. (ii) Hardware agnosticism: operations dispatch to NumPy or CuPy. (iii) Severity monotonicity: severity 1 is mild and 5 severe, with monotone parameters (Table 2), enabling the Spearman-ρ analysis of Section 4.

3.2.1 Physical mechanisms: how each corruption arises in hardware

The six corruptions are not arbitrary perturbations; each models a documented event-camera failure, which is what makes Gen1-C a robustness test rather than a synthetic curiosity. Figure 1 contrasts the normal sensor output with the corrupted output for each, and names the mechanism; we expand the physics here.

- hot_pixel – defective / stuck pixels, fixed-pattern noise. Fabrication defects, bias-current mismatch and junction leakage make individual DVS pixels cross their change threshold with no scene change, firing continuously at fixed coordinates. The rate of such self-triggering rises with temperature and with aggressive bias settings, so it materializes as a constant DC offset at a fixed set of locations – the event-camera analogue of stuck CCD/CMOS pixels.
- event_flood – EMI, illumination transients, readout saturation. Electromagnetic interference, LED/PWM flicker, or a sudden strong-light transient drives a dense burst of spurious events in a region for a short window; when the local event rate exceeds the readout/arbiter bandwidth the periphery saturates, so the burst is both localized in space and bounded in time.
- temporal_jitter – clock noise, timestamp error. Row/column arbitration delay, bus contention, and finite timestamp resolution misassign events to neighbouring time bins. The marginal per-pixel counts are essentially preserved; what is corrupted is the order (and hence the temporal autocorrelation) of events.
- event_rate_shift – contrast-threshold / biasing drift, domain gap. The per-pixel contrast threshold and bias currents set the event rate; temperature drift, re-biasing, or simply moving between a low-texture/slow-motion and a high-texture/fast-motion regime multiplies the global rate. We model both directions (under- and over-rate) with a seeded coin flip.
- polarity_flip – ON/OFF misclassification. Comparator offset and asymmetric ON/OFF thresholds (a documented DVS non-ideality) cause the brightness-change sign to be reported incorrectly for a fraction of events; under fast global illumination swings whole frames can have their polarity inverted.
- spatial_dropout – dead-pixel clusters, occlusion. Manufacturing dead regions, dust/lens contamination, condensation, or partial physical occlusion zero the event support over contiguous areas for the duration of the fault.

3.2.2 Corruption algorithms and severity schedules

Each corruption operates on a histogram $X \in \{0, \ldots, 255\}^{N \times 20 \times H \times W}$ and returns a corrupted copy. Hot pixel: add a constant δ to every bin/frame at n fixed random locations (int16, clipped). Event flood: at b random frame/patch centres add δ inside a patch covering fraction f over frames within ± 2 of the centre. Temporal jitter: roll the ON block (ch. 0–9) and OFF block (ch. 10–19) along the time-bin axis by independent integer shifts in $[-s_{\max}, s_{\max}]$, zero-filling vacated bins. Event-rate shift: multiply all counts by a seeded under/over factor and clip. Polarity flip: swap the ON/OFF blocks on a random fraction p of frames. Spatial dropout: zero k rectangular regions of size (f_h, f_w) across all channels/frames. Severity schedules are in Table 2.

Figure 1: The six Gen1-C corruptions and their physical mechanisms. For each corruption: a normal event frame (left) and the corrupted frame the modelled hardware fault produces (right). hot_pixel adds persistent stuck-pixel firings; event_flood injects a dense local burst; temporal_jitter smears events across time bins; event_rate_shift scales the global density; polarity_flip swaps ON/OFF; spatial_dropout blanks a region. Each maps to a distinct, predictable signature in the membrane moments (Section 3.3).

Corruption	Physical failure mode	Axis of the input perturbed
hot_pixel	defective / stuck pixels	constant DC offset at fixed pixel locations
event_flood	EMI / sensor saturation	localized count bursts in space and time
temporal_jitter	clock noise / timestamp error	time-bin ordering (per polarity)
event_rate_shift	event-density domain gap	global multiplicative rate
polarity_flip	ON/OFF misclassification	polarity sign on a fraction of frames
spatial_dropout	dead-pixel clusters / occlusion	spatial support (rectangular regions zeroed)

Table 1: The six corruptions, the sensor failure each models, and the input axis it perturbs.

Corruption	Sev. 1	Sev. 2	Sev. 3	Sev. 4	Sev. 5
hot_pixel (n, δ)	$10, 20$	$30, 40$	$80, 80$	$150, 140$	$300, 200$
event_flood (b, f, δ)	$2, .05, 60$	$4, .08, 100$	$8, .12, 150$	$15, .18, 200$	$25, .25, 255$
temporal_jitter $s_{\max}$	1	2	3	5	8
event_rate_shift (u, o)	$.80, 1.20$	$.65, 1.40$	$.50, 1.65$	$.35, 2.00$	$.20, 2.50$
polarity_flip p	.05	.10	.20	.35	.50
spatial_dropout (k, f_h, f_w)	$1, .05, .05$	$1, .10, .10$	$2, .10, .10$	$2, .15, .15$	$3, .20, .20$

Table 2: Severity schedules. Symbols: n hot pixels, δ injected count, b burst centres, f patch fraction, $s_{\max}$ max bin shift, (u, o) under/over rate factors, p flipped-frame fraction, k dropout regions, (f_h, f_w) region height/width fractions.

3.3 Why membrane statistics detect corruption: a leaky-integrator theory

We show that the detectability of each corruption is a closed-form consequence of the PLIF state equation, not an empirical accident. The membrane is a leaky exponential accumulator of the input current; each corruption perturbs that current in a structurally distinct way, and the perturbation propagates to a predictable change in the moments of φ.

State equation and closed form. SpikingJelly's MultiStepParametricLIFNode (Fang et al., 2021) updates, per channel, $V[t] = (1 - \frac{1}{\tau})V[t-1] + WX[t]$, emitting a spike with reset when V crosses threshold θ. Writing $\lambda \equiv 1 - 1/\tau \in (0, 1)$ and unrolling from $V[-1] = 0$,

$$V[t] = \sum_{k=0}^{t} \lambda^k \, W \, X[t-k], \tag{2}$$

so V is a causal, leaky linear filter of the event stream with memory horizon $\sim \tau$. Two consequences: V carries the temporal structure of recent input (so temporally-structured corruptions leave a signature even when the marginal input is unchanged), and because $X \to V$ is linear, the moments of V inherit closed-form dependence on the moments of X.

Steady-state moments. For wide-sense-stationary input with mean m, variance s^2, autocovariance $\gamma(j)$,

$$\mathbb{E}[V] = \frac{Wm}{1-\lambda} = \tau W m, \qquad \mathrm{Var}[V] = \frac{(Ws)^2}{1-\lambda^2} + 2W^2 \sum_{j \ge 1} \gamma(j) \frac{\lambda^j}{1-\lambda^2}. \tag{3}$$

The mean tracks the input rate amplified by τ; the variance tracks the input variance plus an autocovariance term that depends on the time ordering, not the marginal. These are the levers each corruption pulls.

Per-corruption predictions. hot_pixel adds a constant δ, shifting $\mathbb{E}[V]$ by $\tau W \delta$ until V saturates at θ: affected μ_c separate from the clean range, giving inevitable detection (AUROC ≈ 1, monotone). temporal_jitter is a measure-preserving permutation: it leaves m, s^2 (hence $\mathbb{E}[V]$ and the first variance term) unchanged but perturbs $\gamma(j)$ and the lag-1 autocorrelation of $V(t)$. The deployed MDD uses only the static moments $\varphi = [\mu, \sigma^2, \kappa]$ (lag-1 autocorrelation is named here as the mechanism, not a feature we compute); jitter surfaces in φ because the deeper layers' larger effective τ convert this timing change into a σ^2 shift – which is why the layer-4 Mahalanobis branch, not any temporal statistic, is the strongest single view. event_rate_shift scales $m \to \alpha m$, displacing every channel's mean along one global-activity direction – detectable by a 1-D projection, with a phase transition once α leaves the clean band. spatial_dropout lowers the spatial input variance, hence σ_c^2, while μ_c stays intact; but GAP pools away the spatial dimension before the moments are taken, so the surviving signal makes the membrane look quieter and more typical – under one-sided distance scoring the detector inverts (AUROC < 0.5). event_flood raises m nearly uniformly, the same signature a genuinely busy clean scene produces, giving near-chance per frame; what distinguishes it (spatially unstructured activity; a temporally consistent bias) lives in the spatial dispersion GAP removes and in sequence-level consistency. polarity_flip: the trained detector is approximately polarity-symmetric, so WX – and thus φ – is near-invariant; the AUROC ceiling is ≈ 0.5 by construction.

Taxonomy. This yields a falsifiable taxonomy that organizes the whole results section: magnitude corruptions (hot_pixel, event_rate_shift, jitter-via-variance) move φ away from clean and are caught by distance detectors; structure corruptions (spatial_dropout, the spatial part of event_flood) leave the marginal moments intact and require the spatial-dispersion features GAP discards or a rule that flags "too typical"; invariant corruptions (polarity_flip) are near-orthogonal to the membrane and are an input-side problem. Every prediction follows from (2)–(3) with no parameters fit to corruption labels.

3.4 Severity monotonicity: three guarantees and the spike–membrane gap

The predictions above are qualitative. We now make them quantitative: under mild assumptions the membrane statistics, and the OOD score built on them, move monotonically with corruption severity. Let severity be $\gamma \geq 0$ and model a corruption family as an additive drift on the per-layer input current, $X_t(\gamma) = X_t(0) + \gamma\Delta_t + \epsilon_t(\gamma)$, with $\mathbb{E}[\epsilon_t] = 0$. Writing the membrane deviation $\delta V_t(\gamma) = V_t(\gamma) - V_t(0)$ and linearising the PLIF update $V[t] = (1 - \frac{1}{\tau})V[t-1] + WX[t]$ in the sub-threshold regime (bounded reset mismatch) gives $\delta V_t(\gamma) = \lambda\, \delta V_{t-1}(\gamma) + W(\gamma\Delta_t + \epsilon_t)$, which unrolls to $\delta V_t(\gamma) = \sum_{k=0}^{t} \lambda^{\,t-k} W(\gamma\Delta_k + \epsilon_k)$.

Theorem 1 (Monotone membrane drift). If the corruption induces a non-zero current drift $\mathbb{E}[\Delta_k] \neq 0$ in at least one monitored PLIF layer and the reset mismatch is bounded, then the temporal mean of the membrane trajectory shifts at least linearly in γ, $\mathbb{E}[\delta V_t(\gamma)] = \gamma \sum_{k=0}^{t} \lambda^{\,t-k} W\, \mathbb{E}[\Delta_k]$, up to a bounded reset term. Hence the membrane mean coordinate μ of φ is OOD-informative.

Theorem 2 (Monotone membrane variance). Suppose the corruption inflates the input-current covariance, $\mathrm{Cov}(X_t(\gamma)) = \Sigma_0 + \gamma\Sigma_\Delta$ with $\Sigma_\Delta \succeq 0$. Then for the stable linearised PLIF recurrence the stationary membrane covariance is $\Sigma_V(\gamma) = \big(\Sigma_0 + \gamma\Sigma_\Delta\big)\, W^2/(1-\lambda^2)$, so $\frac{d}{d\gamma} \operatorname{tr} \Sigma_V(\gamma) = \frac{W^2 \operatorname{tr}(\Sigma_\Delta)}{1-\lambda^2} \geq 0$. Any corruption that increases the (PSD) input covariance therefore monotonically increases membrane variance — justifying the σ^2 and κ coordinates of φ, not only the mean.

Theorem 3 (MP-OOD score monotonicity). Let $\mathbb{E}[\varphi_\gamma] = m_0 + \gamma b$ and $\mathrm{Cov}(\varphi_\gamma) = C_0 + \gamma C_\Delta$ with $C_\Delta \succeq 0$, where (m_0, C_0) is the clean calibration model. The expected Mahalanobis OOD score obeys

$$\mathbb{E}\big[D^2(\varphi_\gamma)\big] = d + \gamma \operatorname{tr}(C_0^{-1} C_\Delta) + \gamma^2\, b^\top C_0^{-1} b, \tag{4}$$

which is non-decreasing in γ and strictly increasing whenever the feature mean shifts ($b \neq 0$) or its covariance grows ($C_\Delta \neq 0$).

Theorem 3 is the formal basis of the severity-monotonicity we measure (Spearman $\rho = +1$ for hot_pixel, event_rate_shift, event_flood; Appendix H). The lone exception, spatial_dropout ($\rho = -1$), is predicted: GAP removes the dimension in which the shift b lives, so the score sees only a contraction toward m_0 and the sign of (4) flips — the inversion of Section 3.3 is exactly the failure of the theorem's premise when the discriminative coordinate is pooled away.

Proposition 1 (Spike counts discard sub-threshold OOD shift). Consider a neuron with threshold θ whose in-distribution membrane lies in $[0, \theta - \varepsilon]$, and a corruption that shifts every membrane value by $0 < \delta < \varepsilon$. Then the spike train is identically zero under both distributions ($s_t^{\mathrm{ID}} = s_t^{\mathrm{OOD}} = 0$ for all t), so a spike-count detector cannot separate them; yet the membrane means differ by exactly $\bar{V}^{\mathrm{OOD}} - \bar{V}^{\mathrm{ID}} = \delta$.

Proposition 1 is the conceptual novelty over prior spike-count SNN-OOD (Martínez et al., 2023): spike-count OOD detects only threshold-crossing distribution shift, whereas membrane-potential OOD detects both threshold-crossing and sub-threshold shift, and so strictly subsumes it. The sub-threshold potential is the richer signal precisely because spiking is a lossy thresholded projection of it.

3.5 The neuromorphic hardware advantage

We are careful to separate two cost claims. The signal φ is zero added inference compute: on Loihi (Davies et al., 2018), BrainScaleS (Schemmel et al., 2010), and SpiNNaker (Furber et al., 2014) the membrane potential V is physical state the substrate already maintains, so producing φ is an $O(\text{channels})$ register read plus three moments per channel – zero MACs, zero weight fetches, no second network. The detector (MDD) is not claimed to be free: it operates on the 2112-D vector φ off the critical path (a host or co-processor), with PCA projection, a Mahalanobis form, and a cosine-kNN for RCF. Crucially, that cost is bounded by the size of φ – a few thousand floats per frame – and involves no second forward pass over the input, the very thing every ANN OOD detector requires (Table 3). So the comparison is not "φ beats ANN baselines on a GPU" but "φ supplies an OOD signal within the SNN inference budget, with a lightweight downstream test, whereas the ANN detectors need a full non-spiking forward pass the deployment never runs."

We do not claim φ is the only on-chip-feasible OOD signal: simple spike-rate or spike-entropy statistics are equally cheap to read. We claim it is the richest such signal – it provably subsumes spike counts (Proposition 1) – and the only one that, unlike the ANN families below, needs no logits, gradients, or re-forwarding. We measure φ in simulation (SpikingJelly on GPU); the inference-cost claim is architectural (it follows from V being a maintained register), and an on-silicon energy/latency measurement of the full pipeline is future work.

3.6 The Manifold-Decomposition Detector (MDD)

The theory classifies corruptions by how they move φ relative to the clean manifold: dilations push outward (hot_pixel, event_rate_shift; jitter in the deep-layer subspace), contractions pull toward the mode (spatial_dropout, event_flood) and thereby invert one-sided distance/density detectors, and invariant corruptions (polarity_flip) leave no membrane signature. No single signed scalar can face outward, inward, and into a subspace at once – which is why every prior single detector (Mahalanobis, k-NN, GMM, flow, learned fusion) topped out near 0.70 macro and inverted on contractions. MDD stops collapsing φ to one number and scores three (optionally four) orthogonal, two-sided axes, fused by a calibrated OR:

- B1 radius (two-sided isotropic energy): $|\,\|z\| - \mathbb{E}\|z\||/\text{sd}$ in a PCA-64 space – the dilation/contraction energy axis.
- B2 RCF (Radial-Conditional, the novel core): a direction-conditioned radius $|\,\|z\| - \mathbb{E}[\|z\| \mid \text{dir}]|/\text{sd}$ via cosine-kNN; flags under-dispersion – the first branch to take contractions above chance per frame.
- B3 L4 Mahalanobis (one-sided) on the layer-4 $[\mu, \sigma^2, \kappa]$ block – the subspace axis for temporal_jitter.
- B4 spatial (when phi_spatial is extracted): Ledoit–Wolf Mahalanobis on the spatial-dispersion features.

Each branch is standardized on held-out clean and fused by a calibrated max. The shipped default folds an empirical-covariance Mahalanobis into B1 (radius′) and replaces the plain max with a studentized combiner $\max - \alpha \cdot \text{median}$ ($\alpha \approx 0.5$); the derivation and ablation are in Appendices C–E. The detector is fit once on clean-train only and is corruption-blind; an optional per-recording aggregation integrates the consistent per-sequence bias.

4 Experimental Results

4.1 Setup

Prophesee Gen1 test split: 470 sequences, $\approx$ 343,099 frames; each corruption applied per whole sequence at L1–L5. Detectors are fit on clean-train frames only and are corruption-blind; the split honours seq_lens so no clean frame appears in both fit and eval (vmem_utils.split_train_eval/held_out_eval). We report AUROC (and, in the appendix, AUPR / FPR@95%) per-frame and per-sequence. All AUROC values are oracle-labelled separability point estimates on a single split; bootstrap CIs are not yet banked (Section 5). Extraction is strictly $B = 1$ (SpikingJelly's batch axis is the time axis).

4.2 Motivation: a SOTA detector collapses under event corruptions

We take the published, pretrained hybrid SNN–ANN detector (Neftci et al., 2025) (clean Gen1 test mAP ≈ 0.36) off the shelf and feed it corrupted input with no retraining, intercepting the event histogram after loading and before the backbone and applying each Gen1-C corruption at each severity. mAP is computed by the model's native Prophesee evaluator, so the only change versus a stock validation run is the corruption of the input tensor. Reproducible from HybridDetection/validation_corrupt.py; Table 4 reports the result.

4.3 Main result: MDD OOD detection

Tables 5–6 report fused MDD AUROC per corruption/severity, per-frame and per-sequence (source: outputs/results/final_results.csv). One unsupervised, corruption-blind score takes four of six corruptions to ≥ 0.95 AUROC per sequence at severity 5 (hot_pixel and event_flood reach 1.00); sequence aggregation rescues event_flood from 0.55 per-frame to **1.00** (the strongest frame$\rightarrow$sequence jump) and lifts event_rate_shift to 0.955 and temporal_jitter to 0.952. The two residuals are theory-predicted dead ends.

What aggregation does (and does not) buy. Because each Gen1-C corruption is applied to the whole sequence, every frame of a corrupted recording carries the same bias; averaging the per-frame score over a recording integrates that consistent bias and separates the classes at a rate $\sim \sqrt{\#\text{frames}}$. This is why a near-chance per-frame signal (event_flood, 0.551) becomes perfectly separable per sequence (1.000): the lift is real but partly a property of the evaluation protocol, not evidence that any single frame is detectable. It would shrink if only a fraction of frames were corrupted, or if the decision had to be taken per frame. We therefore report per-frame AUROC as the conservative signal and treat per-sequence AUROC as an optimistic bound at a coarser (per-recording) decision granularity; the window sweep of Section 4.6 makes this dependence explicit, separating corruptions genuinely solved per frame from those solved only by pooling.

Severity 2 was not extracted; severities $1, 3, 4, 5$ are shown. Numbers are the shipped improved-MDD default (radius$'$ + studentized fusion), lifted verbatim from outputs/results/final_results.csv (plot_corruption_graphs.py).

4.4 The geometry of corruption

The AUROC tables say how well each corruption is detected; Figure 3 shows why, making the dilation/contraction/invariant taxonomy visible. Membrane energy $\|z\|$ moves right for dilations and left for contractions (Figure 3a); the Mahalanobis d^2 of the contractions sits at smaller values than clean (Figure 3b), which is exactly the AUROC< 0.5 inversion; and the clean$\rightarrow$corrupt centroid drift in a shared clean-PCA (Figure 3c) separates the corruption directions, with event_rate_shift and spatial_dropout sharing an axis (Appendix M, Figure 9).

4.5 Branch decomposition and fusion ablation

Figure 4a shows each branch is corruption-specialized (radius owns dilations, L4 owns timing, RCF/spatial own contractions); the plain calibrated max sits below the best single branch on both residuals, motivating the studentized fusion (Appendix D). Folding an empirical-covariance Mahalanobis into the radius branch and replacing max with $\max - \alpha \cdot \text{median}$ (Figure 4b,c) recovers spatial_dropout from 0.560 toward its RCF ceiling (0.689) and lifts the worst-corruption floor, with no easy-corruption regression. The α knob trades polarity vs. dropout (Figure 4c).

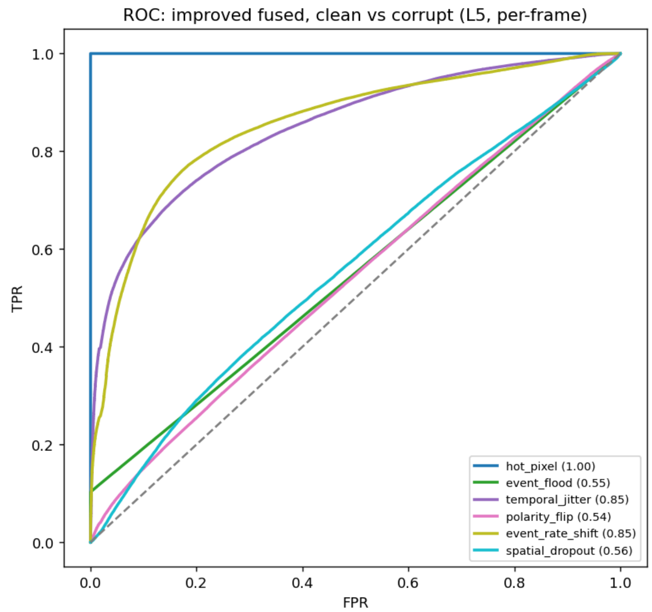
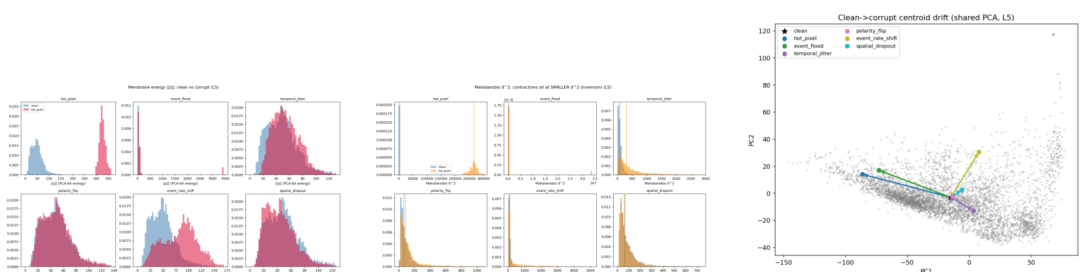


Figure 2: ROC curves of the improved fused MDD at L5 (per-frame), clean vs. each corruption. Source: plot_corruption_graphs.py (18_roc_curves_L5).



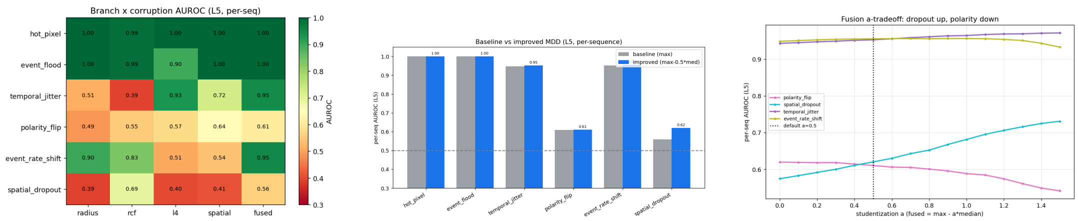
(a) radial energy $\|z\|$ (b) Mahalanobis d^2 (inversion) (c) PCA drift arrows

Figure 3: Geometry of corruption (L5, leakage-safe). (a) Dilations move energy right, contractions left, polarity overlaps clean. (b) Contractions sit at smaller d^2 than clean – the source of AUROC< 0.5. (c) Clean→corrupt centroid drift in a shared clean-PCA. Source: plot_corruption_graphs.py.

4.6 Windowed aggregation: the deployable operating point

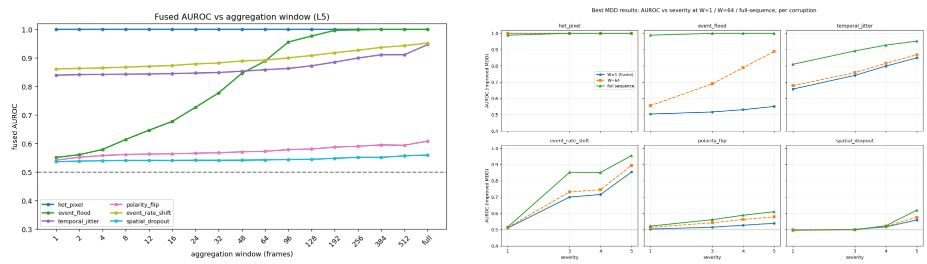
Full-sequence aggregation (Table 6) decides once per recording ($\sim$730 frames) and is therefore an optimistic bound. The practically relevant – and more reliable – setting is a short, fixed window of W frames, which bounds decision latency. Table 7 sweeps W at severity 5 (source: outputs/results/mdd_window_sweep.csv). A modest window already recovers most of the signal without the whole-recording assumption: at $W = 64$, event_rate_shift reaches 0.89, temporal_jitter 0.86, event_flood 0.89 and hot_pixel 1.00; by $W = 128$ event_flood is at 0.98. We therefore report $W \in [32, 64]$ as the headline deployable operating point – strong on the four solved corruptions at a bounded latency, rather than the once-per-recording full-sequence number.

Which results are manufactured by pooling. The window curve separates genuine from manufactured gains. event_flood has almost no per-frame signal and only becomes separable as W grows ($0.55 \to 0.78 \to 0.89 \to 0.98$): real, but a property of integrating a temporally-consistent bias, so we report the windowed value rather than the trivial full-



(a) branch × corruption (b) baseline vs. improved (c) α trade-off

Figure 4: Branches and fusion (per-sequence, L5). (a) Per-branch AUROC heatmap. (b) Baseline max vs. improved (radius′ + studentized fusion). (c) AUROC vs. studentization α: dropout↑, polarity↓. Source: plot_corruption_graphs.py. The combiner sweep is Figure 7.



(a) fused AUROC vs. window (b) best AUROC vs. severity

Figure 5: Aggregation and severity. (a) Fused AUROC vs. pooling window per corruption. (b) Improved-MDD AUROC vs. severity at $W = 1$ / 64 / full sequence. Source: plot_corruption_graphs.py.

recording 1.00. hot_pixel is genuinely solved at every window. polarity_flip (0.54→0.59) and spatial_dropout (0.54→0.56) stay flat across all windows – pooling cannot rescue what is not present per frame, which is precisely why these two are the honest hard cases.

4.7 Discussion: physical causes and biological adaptation

The value of the closed-form theory is that each physical axis (Section 3.2.1) maps to a distinct statistical axis of the membrane moments, so a detector's behaviour on a corruption is predictable from the sensor failure it models. One mechanism is worth dwelling on. event_flood – one of the corruptions our membrane detector handles worst per frame (near chance, Table 5) – inflates every channel's mean proportionally, mimicking a busy scene (Section 3.3). Biological neurons do not integrate a flood passively: spike-frequency adaptation (calcium- and slow-K$^+$-driven) (Benda & Herz, 2003; Gabbiani & Krapp, 2014) and short-term synaptic depression (a divisive gain control) (Abbott et al., 1997; Rothman et al., 2009) normalise sustained drive so the neuron responds to changes, not absolute level. The PLIF neuron has none of this – a single static leak τ, a linear accumulator (Eq. 2), no adaptation – so a flood simply scales the accumulator and looks like a busy scene. The corruption biology handles best is the one our biologically-simplified model handles worst; equipping the front-end with adaptive (SFA / AdLIF) neurons is a concrete, biologically-grounded path to both detecting event_flood and restoring mAP (future work, cf. Section 4.2).

5 Limitations and the Open Challenge

Honest residuals. Two corruptions remain unsolved from the membrane alone, and the theory predicts why. spatial_dropout's signature is spatial and GAP discards spatial layout; the released spatial-dispersion features (phi_spatial) feed MDD's spatial branch, which we validated end-to-end on a controlled contraction (a signal present only in phi_spatial is lifted from fused-without-spatial AUROC ≈ 0.50 to ≈ 1.00; Appendix L). polarity_flip is a property of the trained network's polarity invariance, not of our representation, and is undetectable from V_{mem} by construction. We frame both as an open challenge: the Achilles' heel of event-based perception, where corruptions are at once harmful (Section 4.2) and hard to detect cheaply.

Other limitations. (i) The zero-cost claim is architectural and applies to the signal only: φ is free at inference, but the MDD detector (PCA/Mahalanobis/cosine-kNN) runs host-side on the 2112-D vector; we measure in simulation (SpikingJelly on GPU) and an on-silicon energy/latency measurement of the full pipeline is future work. (i′) φ is read from the reset-inclusive membrane register, so the linear sub-threshold theory is an approximation; a strictly spike-excluded V and a reset-aware analysis are untested. (ii) AUROC are single-split point estimates without bootstrap CIs, and we do not yet report sensitivity to the PCA dimension, the cosine-kNN k, or the clean-calibration sample size. (iii) Single dataset (Gen1); DSEC/Gen4 transfer is untested. (iv) Severities are synthetic monotone transforms, not measured sensor failures – the contribution is the benchmark/protocol/feature release. (v) Per-sequence AUROC assumes a corruption is consistent across a whole recording (as in Gen1-C); the strong per-sequence numbers (e.g. event_flood 1.000) are partly a consequence of that protocol and of the coarser decision granularity, and should not be read as per-frame detectability. (vi) AUROC is an oracle-labelled separability metric, not a value the corruption-blind detector reproduces at a fixed operating threshold in deployment.

6 Conclusion

We introduced OOD detection for event-camera spiking detectors from the free sub-threshold membrane potential, the Gen1-C corruption benchmark with its physical-mechanism grounding, a closed-form theory of per-corruption detectability, and the Manifold-Decomposition Detector that follows from a dilation/contraction diagnosis. A pre-trained SOTA detector collapses under these corruptions; φ is a zero-added-inference-cost OOD signal that needs no second forward pass and provably subsumes spike-rate signals; and one unsupervised score takes four of six corruptions to ≥ 0.95 with sequence aggregation. The two residuals are theory-predicted and released as an open challenge, with a concrete, biologically-grounded path – spatial-dispersion features and adaptive neurons – toward closing them.

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Appendix

This appendix documents the complete experimental program – including the levers that did not work and the one result we retracted – so the headline claims can be audited against everything we tried. Every subsection names the generating script and source artifact.

A Implementation, Splits, and Reproducibility

Feature extraction. A VmemMonitor hooks all four MultiStepParametricLIFNode layers (64/128/256/256, 704 channels), captures $V \in \mathbb{R}^{T\times1\times C\times H\times W}$, GAPs over space, and reduces to $[\mu, \sigma^2, \kappa]$ per channel (φ, 2112-D). phi_spatial (1408-D) adds per-channel spatial variance and participation ratio, stored float32 (float16 overflows spatial_var under event_flood).

Constraint BATCH_SIZE$=$ 1. SpikingJelly's multi-step PLIF treats the batch axis as time; $B > 1$ leaks membrane state across sequences. Throughput scales by parallel workers (run_parallel_extract.py).

Protocol. Leakage-safe fit on clean-train only, corruption-blind; the split honours seq_lens. Metrics: AUROC, AUPR, FPR@95% (evaluate_detectors.py), per-frame and per-sequence. No bootstrap CIs yet. Config constants (benchmark_config.py): BATCH_SIZE=1, PHI_SAVE_EVERY=5, PLIF_LAYERS=None, TRAJ_SAVE_N=50.

B Baseline OOD Detectors

We fit seven post-hoc detectors on clean φ (fit_detectors.py) and score them on every corrupted run (evaluate_detectors.py): Mahalanobis (Ledoit–Wolf; the reference detector), k-NN ($k = 5$), GMM (5), OCSVM (RBF), PCA (50), MLP-AE, RealNVP. Table 8 gives the reference behaviour that frames the benchmark: three corruptions sit below chance (informative but inverted – corrupted lies closer to the clean mean than held-out clean).

ANN baselines. A ResNet-18 on event-image / voxel-grid inputs scored with 16 post-hoc OOD detectors spanning 2017–2024 (evaluate_ann_baselines.py): the established MSP, ODIN, Energy, Mahalanobis, deep-kNN, ReAct, ViM, DICE, GradNorm, plus recent SOTA — MaxLogit (2022), GEN (CVPR'23), ASH (ICLR'23), NNGuide (ICCV'23), SCALE (ICLR'24), fDBD (ICML'24) and NECO (ICLR'24). All are implemented faithfully from the offline feature/logit cache (no retraining, no input gradients, no second model pass beyond the untouched classifier head; input-perturbation / multi-layer-ensemble steps of ODIN and Mahalanobis are dropped, as noted in the code). These ANN detectors are not on-chip-feasible (Section 3.5) and are reported only as an upper reference.

FIGURE/TABLE TODO: Insert the 16-detector ANN-baseline AUROC table once the full grid is run.

C MDD: Per-Branch Decomposition

Table 9 reports each branch in isolation (per-sequence, L5). The plain calibrated max sits below the best single branch on both residuals (spatial 0.639 > 0.609; rcf 0.693 > 0.560) – it inflates the clean floor.

D Residual-Corruption Improvement Loop

Under a strict no-re-extraction, no-new-branch constraint we improved only the existing branches and the fusion rule (wired into analysis/mdd.py as the default, toggleable). (A) Radius covariance enrichment: $\text{radius}' = \max(\|z\|\text{-energy}, \text{emp\_maha64})$, where emp_maha64 is an empirical (unshrunk) covariance Mahalanobis in PCA-64 that catches the covariance rotation a polarity flip induces (polarity 0.656 > spatial 0.639) without the dilation loss of full whitening (0.900 → 0.600 on event_rate_shift). (B) Studentized fusion: $\max - \alpha \cdot \text{median}$ ($\alpha \approx 0.5$) so a lone strong branch survives while diffuse clean noise cancels. Table 10 and Figure 6.

Honest ceiling. Neither residual reaches 0.8: spatial_dropout (contraction) caps ≈ 0.69, polarity_flip (polarity-symmetric network) ≈ 0.66. Breaking 0.8 needs information not in the current φ (temporal $V(t)$, finer spatial pooling, the raw polarity channel) – out of scope under no-re-extraction.

Timeline. Table 11 records the exp1–exp10 trail, including dead ends.

E Fusion-Combiner Ablation

We swept combiners over the calibrated branches: max, mean, top-2 mean, p-norm, quantile-rank, studentized $\max - \alpha\{\text{mean(rest)}, \text{median}, \min\}$. Plain max sits below the best branch on residuals; rank/quantile helps residuals but collapses easy corruptions (event_flood

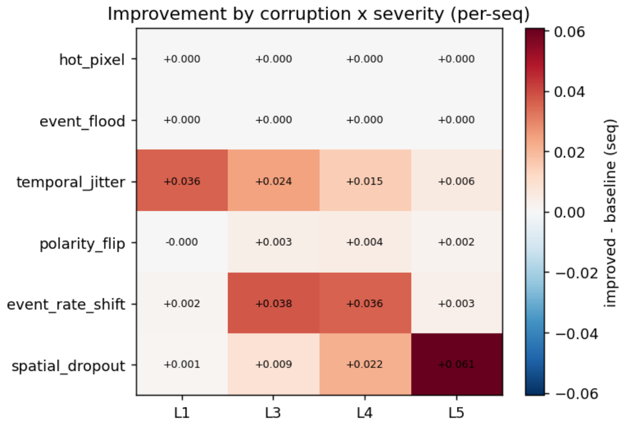


Figure 6: Improved − baseline per-sequence AUROC, per corruption × severity. Source: plot_corruption_graphs.py (22_improvement_by_severity).

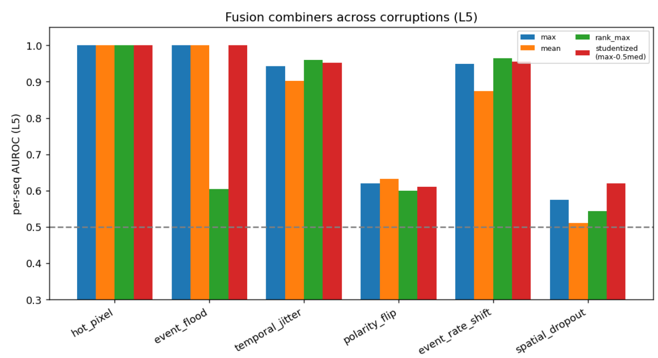


Figure 7: Fusion-combiner sweep across corruptions (per-sequence, L5). Source: plot_corruption_graphs.py (17_fusion_combiners).

$1.00 \to 0.61$); studentized max $-\alpha$ median is the Pareto winner (Figure 7). Supervised LR fusion buys nothing per-frame and fails to generalize (LR leave-one-out ≈ 0.49).

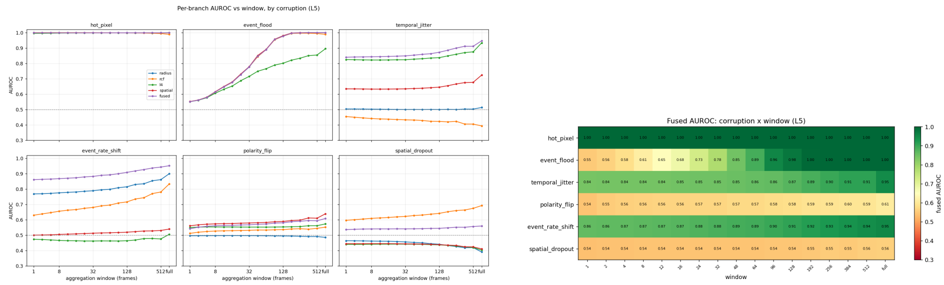
F Severity and Window Sweep (full)

Figure 8 gives the per-branch window sweep and the corruption×window heatmap (companion to Figure 5). The fused-vs-best-branch (oracle) gap is in paper_figures.md; the headline gap is event_flood's per-frame→windowed→per-sequence climb ($0.55 \to 0.89$ at $W{=}64 \to 1.00$).

G Representation Ablation

Is the sub-threshold membrane the right signal? representation_ablation.py scores membrane φ (full_membrane) and its $\mu/\sigma^2/\kappa$ and per-layer slices (L4 carries jitter); spike-rate / spike-entropy; ANN feature (last_ann_gap) and logits (head_cls_L0_gap); and fused (concat + LogReg).

FIGURE/TABLE TODO: Insert the representation-comparison AUROC table from representation_ablation.py; PENDING the full-grid run.



(a) per-branch vs. window

(b) corruption × window

Figure 8: Window sweep detail. Source: plot_corruption_graphs.py.

H Severity Monotonicity, Cross-Corruption, and What φ Encodes

Spearman ρ. severity.py: hot_pixel $\rho = +1$, spatial_dropout $\rho = -1$ (anti-detectability).

FIGURE/TABLE TODO: Insert the per-corruption Spearman-ρ table.

Cross-corruption zero-shot. cross_corruption.py holds out one corruption type and evaluates on it.

FIGURE/TABLE TODO: Insert the cross-corruption transfer matrix.

Beyond binary OOD. analyse_comparisons.py trains, on φ, a 7-class corruption-type classifier (confusion matrix) and a severity regression (R^2).

FIGURE/TABLE TODO: Insert the confusion matrix and severity-regression R^2 table.

I Temporal and Sequence Representations (and a Retracted Result)

We evaluated handcrafted temporal_phi (28-D), a Temporal-AE (1D-CNN on the GAP'd $V(t)$ trajectory), and margin histograms ($V - \theta$ quantized).

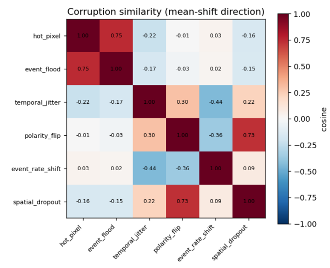
Retraction (honest negative). An earlier internal finding claimed temporal features rescue the hard corruptions to ≈ 0.85 and that "the paper's strongest version is a temporal paper." This does not reproduce. It was computed on the 50-sample raw-trajectory path (35 train/15 test from the first 50 frames of one sequence, a 28-D Mahalanobis on 35 autocorrelated points, without a held-out clean split). Re-running the exact method on freshly extracted data gives 0.29/0.04/0.06 – the 0.85 was leakage plus 50-sample noise (a refactor: unified train/eval split commit postdates the finding).

Honest picture. With a leakage-safe split, handcrafted temporal beats static only modestly: event_flood ≈ 0.625 (static ≈ 0.58), spatial_dropout ≈ 0.541 (static ≈ 0.47) – a real ~ 0.05–0.08 gain, not a rescue. The Temporal-AE is the weakest detector. The genuine contraction lift comes from sequence aggregation (Section 4.6).

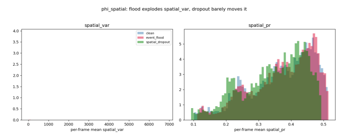
J Levers Tried (Negative-Results Catalogue)

K Free-Rider Validity Check

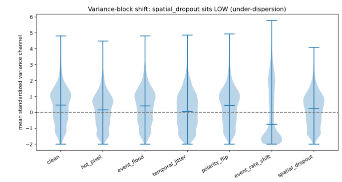
free_rider_ablation.py compares OOD AUROC from (A) trained-SNN membrane, (B) random-init SNN membrane, (C) raw-input statistics; the premise requires (A)$\gg$(C).



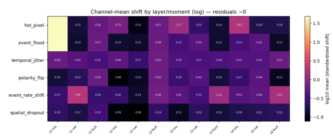
(a) corruption similarity



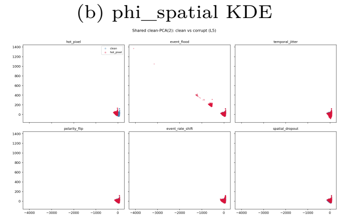
(b) phi_spatial KDE



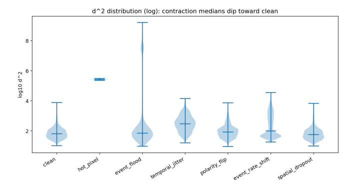
(c) σ^2 shrinkage



(d) channel-shift fingerprint



(e) shared-PCA scatter



(f) log d^2 violins

Figure 9: Additional diagnostics. (a) cosine similarity of mean-shift directions (rate_shift↔dropout cluster); (b) spatial_var/spatial_pr clean vs. flood vs. dropout; (c) variance-block shift (dropout sits low = under-dispersion); (d) per layer/moment standardized channel shift; (e) shared clean-PCA scatter; (f) log d^2 violins (contraction medians dip toward clean). Source: plot_corruption_graphs.py.

FIGURE/TABLE TODO: Insert the trained-vs-random-vs-raw table; PENDING run – a gating experiment.

L Spatial-Branch Mechanism Check

On a controlled contraction whose signal lives only in phi_spatial (GAP'd φ matched to clean): fused-without-spatial AUROC $= 0.496 \rightarrow$ fused-with-spatial $= 1.000$; spatial branch alone $= 1.000$ – confirming the wiring. On the banked extraction the spatial branch nails event_flood (1.000) but stays below chance on spatial_dropout (0.411), consistent with the contraction ceiling.

M Additional Geometry Figures

Figure 9 collects the remaining geometry/diagnostic panels.

N Released Artifacts: Gen1-C

Corruption benchmark (code + seeds): six corruptions $\times$ five severities, fully seeded; a (corruption, severity, seed) triple regenerates the exact stream via apply_corruption(..., rng); NumPy/CuPy-agnostic on the standard $(N, 20, H, W)$ histogram. We ship the library + seed manifest + config that regenerate from the user's own Gen1 copy (not redistributed histograms). Released φ feature set: pre-extracted membrane statistics for clean + 30 corrupted runs (with seq_lens) so detectors can be benchmarked without GPU inference. Protocol: the leakage-safe split, per-frame / per-sequence AUROC, and the 7-detector + MDD harness. Action before release: confirm the Prophesee Gen1 license permits distributing derived features / regeneration code.

Method	Required extra computation	Why infeasible on neuromorphic HW
Vmem-φ (ours)	native V register read + 3 moments (signal); host-side test on a 2112-D vector (detector)	no logits/gradients/re-forwarding; no second forward pass over the input
MSP (Hendrycks & Gimpel, 2017) / Energy (Liu et al., 2020)	softmax / log-sum-exp over logits	cores emit spikes, not float logits; no on-chip softmax
ODIN (Liang et al., 2018)	temperature softmax + input perturbation	perturbation needs an input-gradient pass the chip cannot take
ReAct (Sun et al., 2021)	activation clipping + second forward pass	re-forwarding violates the SNN energy budget
ViM (Wang et al., 2022)	SVD / dense linear algebra on features	no dense float linear-algebra unit; data movement too costly
GradNorm (Huang et al., 2021)	gradient backprop at inference	neuromorphic inference has no backward pass
kNN / Mahalanobis on ANN features (Lee et al., 2018; Sun et al., 2022)	an auxiliary ANN forward pass	requires a second, non-spiking network – defeats the SNN premise

Table 3: Standard OOD detectors each require a capability absent from the spiking, feed-forward, integer-event execution model – chiefly a second, non-spiking forward pass. Vmem-φ needs none: the signal is the neuron state itself, and the downstream test runs on a small per-frame vector. (We do not claim it is the only on-chip signal – spike-rate/entropy are also cheap but carry less information, Prop. 1.)

FIGURE/TABLE TODO: Insert results/neftci_map_degradation_table.tex once the GPU sweep is run. Rows = 6 corruptions, columns = severities 1–5, cells = mAP; clean baseline mAP ≈ 0.36 in the caption. (Modal/A100: needs the Gen1 test set + gen1_mAP36.ckpt.)

Table 4: Detection mAP of the pretrained hybrid detector under each corruption/severity (no retraining). Pending the GPU sweep. Hypotheses (from Section 3.3): hot_pixel/event_flood swamp object evidence; spatial_dropout deletes object regions; polarity_flip inverts the ON/OFF evidence the head was trained on (degrading mAP even though the membrane cannot flag it – a deliberate detectability-vs-harm mismatch).

Corruption	S1	S3	S4	S5
hot_pixel	1.000	1.000	1.000	1.000
temporal_jitter	0.658	0.742	0.799	0.850
event_rate_shift	0.510	0.701	0.716	0.855
spatial_dropout	0.500	0.502	0.517	0.560
event_flood	0.504	0.517	0.531	0.551
polarity_flip	0.504	0.516	0.527	0.540

Table 5: Fused MDD per-frame AUROC (improved default). Contractions and the invariant polarity_flip sit near chance per frame.

Corruption	S1	S3	S4	S5
hot_pixel	0.989	1.000	1.000	1.000
event_flood	0.989	1.000	1.000	1.000
event_rate_shift	0.516	0.853	0.852	0.955
temporal_jitter	0.810	0.892	0.928	0.952
spatial_dropout	0.497	0.500	0.524	0.620
polarity_flip	0.522	0.562	0.589	0.611

Table 6: Fused MDD per-sequence AUROC. spatial_dropout (0.62) and polarity_flip (0.61) remain the residuals.

Corruption	W=1	W=32	W=64	W=128	full
hot_pixel	1.000	1.000	1.000	1.000	1.000
event_rate_shift	0.861	0.882	0.892	0.909	0.943
event_flood	0.551	0.778	0.889	0.977	1.000
temporal_jitter	0.840	0.849	0.859	0.873	0.911
spatial_dropout	0.536	0.541	0.543	0.545	0.557
polarity_flip	0.542	0.568	0.573	0.581	0.594

Table 7: Fused MDD AUROC vs. aggregation window W (frames) at severity 5. $W = 1$ is per-frame, "full" is whole-recording. A bounded window ($W = 32$–64) already solves the four detectable corruptions; spatial_dropout and polarity_flip stay flat at every window – they are the genuine residuals, not a pooling artifact.

Corruption	Static-φ Mahalanobis AUROC (L5)	Behaviour
hot_pixel	1.000	trivial
temporal_jitter	0.709	moderate
event_rate_shift	0.675	moderate; phase transition ∼sev 3
polarity_flip	0.429	below chance – polarity-symmetric
event_flood	0.408	below chance – moments preserved
spatial_dropout	0.286	anti-detectable ($\rho = -1$)

Table 8: Reference detector (Mahalanobis on static φ, full data, L5). Source: performance_brief.md; generator evaluate_detectors.py.

Corruption	radius	rcf	l4	spatial	fused (max)
hot_pixel	1.000	0.989	1.000	1.000	**1.000**
event_rate_shift	0.900	0.834	0.506	0.540	**0.952**
temporal_jitter	0.513	0.393	0.933	0.724	**0.947**
event_flood	1.000	0.989	0.896	1.000	**1.000**
polarity_flip	0.486	0.552	0.573	0.639	**0.609**
spatial_dropout	0.391	0.693	0.403	0.411	**0.560**

Table 9: Per-branch and fused per-sequence AUROC at L5. Source: mdd_improvements.md §1; harness verified against evaluate_mdd_windows.py.

Corruption	baseline (max)	radius′ + (max −0.5 med)	+(max −1.25 med)
hot_pixel	1.000	1.000	1.000
event_rate_shift	0.952	0.956	0.948
temporal_jitter	0.947	0.952	0.966
event_flood	1.000	1.000	1.000
polarity_flip	0.609	0.607	0.560
spatial_dropout	0.560	**0.613**	**0.689**
mean	0.845	0.855	0.860
min (worst-corruption floor)	0.560	**0.607**	0.560

Table 10: Effect of the two modifications (per-sequence AUROC, L5). Source: mdd_improvements.md §3.

#	Finding
exp1	broad survey: flagged under-dispersion (dropout) + kurtosis (polarity) as the live signals
exp2	faithful harness: rcf 0.693 (dropout), spatial 0.639 (polarity) best singles; fused max dilutes
exp3	selective fusions: rcf+shrink-var → 0.704; spatial+kurt+L1 → 0.662; L1 retains polarity (0.630)
exp4	density inversion / sp_pr / rcf tuning: contraction confirmed but ≤ 0.56
exp5	broaden l4: polarity 0.573 → 0.634, flood-l4 0.896 → 1.000; fused flat ⇒ fusion is the bottleneck
exp6	studentized max−rest: dropout 0.560 → 0.605, floor up
exp7	α-sweep max $-\alpha$ mean(rest): Pareto tension (dropout↑, polarity↓)
exp8	robust floor: max $-\alpha$ median best; $\alpha = 1.5$ ⇒ dropout 0.693
exp9	empirical-cov PCA-64 Mahalanobis = 0.656 (best polarity branch)
exp10	fold emp_maha into radius + studentized-median: both improved, floor 0.560 → 0.607

Table 11: Residual-corruption experiment timeline. Source: mdd_improvements.md §4.

Lever	Result
Two-sided / folded Mahalanobis	rescues the 3 below-chance above chance (dropout 0.286 → 0.531, flood 0.408 → 0.577, polarity 0.429 → 0.547) but degrades jitter (0.709 → 0.615)
Global activity scalar (two-sided σ^2 energy)	big win for event_rate_shift (0.675 → 0.850); near-useless elsewhere
More fit samples (3k→50k)	free but small: jitter 0.709 → 0.757; rest flat
Per-feature standardization before Ledoit–Wolf	jitter 0.709 → 0.820 but hurts event_rate_shift (0.675 → 0.568)
Naive max fusion (two-sided + activity)	worse than either alone
Temporal AE (1D-CNN)	weak everywhere (0.40–0.58)
Handcrafted temporal (leakage-safe)	modest win on flood/dropout (≈ 0.54–0.62)
Supervised LR meta-fusion	no per-frame gain; LOO ≈ 0.49

Table 12: Levers tried on the hard corruptions. Oracle best-achievable per corruption: hot_pixel 1.00, event_rate_shift 0.85, temporal_jitter 0.82–0.84, event_flood 0.63 (frame), polarity_flip 0.55, spatial_dropout 0.54 (frame). Source: performance_brief.md §6.